

Learn IN Depth

Be Professional In Embedded System



Project **2** Learn in Depth

STUDENT MANAGEMENT SYSTEM

Mohamed Fathalla Ahmed | Mastering
Embedded Systems | 16.08.2024

Table of Contents

Project Scope / Problem Statement 2

Methodology / Approach..... 2

Functions Description 3

Software Testing Results..... 6

Project Scope / Problem Statement

For this project, a student information management system software has to be implemented using queue method.

The following operations have to be included in the system:

1. Store the first name of the student
2. Store the last name of the student
3. Store the unique roll number of the student
4. Store the GPA of every student
5. Store the courses registered by the student

Methodology / Approach

The main idea is to develop different functions for each operation. All functions together will form the entire software. As mentioned earlier, **the mechanism of queue data structure (as a dynamic linked list)** is the underlying methodology behind the entire software.

The entire SW is built up based on the following files:

- Queue_LinkedList.c
- Queue_LinkedList.h
- main.c
- student_data_file.txt

More details of the functions will follow in the next chapters.

Functions Description

Main.c is including the main infinite loop with all options to be selected by the user of the software. As well as, **Queue_LinkedList.h** is included which provides all functions needed for the software.

```
/*
#include "Queue_LinkedList.h"

int main (void) {

    char temp_text[40];
    student_queue = createQueue_student();
    DPRINTF("Welcome to the Student Management System \n");
    while (1)
    {
        DPRINTF("Choose The Task that you want to perform \n");
        DPRINTF("1. Add the Student Details Manually \n");
        DPRINTF("2. Add the Student Details From Text File \n");
        DPRINTF("3. Find the Student Details by Roll Number \n");
        DPRINTF("4. Find the Student Details by First Name \n");
        DPRINTF("5. Find the Student Details by Course ID \n");
        DPRINTF("6. Find the Total Number of Students \n");
        DPRINTF("7. Delete the Student Details by Roll Number \n");
        DPRINTF("8. Update the Student Details by Roll Number \n");
        DPRINTF("9. Show all information \n");
        DPRINTF("10. To Exit \n");
        DPRINTF("Enter your choice to perform the task: \n");
        gets(temp_text);

        switch(atoi(temp_text))
        {
            case 1:
                add_student_manually();
                break;

            case 2:
                add_student_file();
                break;

            case 3:
                find_rl();
                break;

            case 4:
                find_fn();
                break;

            case 5:
                find_c();
                break;

            case 6:
                tot_s();
                break;

            case 7:
                del_s();
                break;

            case 8:
                up_s();
                break;

            case 9:
                show_s();
                break;

            case 10:
                DPRINTF("\n Session closed, See you later :) \n\n");
                return 0;

            default:
                DPRINTF("\n Wrong Option: Try Again \n\n");
                break;
        }
    }

    return 0;
}
```

In the “**Queue_LinkedList.h**” the queue structure is defined along with the declarations of the functions needed to handle the queue mechanism and also the user functions:

```
#ifndef QUEUE_LINKEDLIST_H_
#define QUEUE_LINKEDLIST_H_

#include "stdio.h"
#include "stdlib.h"
#include "string.h"

#define DPRINTF(...)      {fflush(stdout);fflush(stdin);\
                           printf(__VA_ARGS__);\
                           fflush(stdout);fflush(stdin);}

// Node structure for the linked list
struct sinfo{
    char    fname[50];
    char    lname[50];
    int     roll;
    float   GPA;
    int     cid[10];
};
// Queue Student Node (Data items (Student Data) + pointer to next element/node )
typedef struct SStudentNode{
    struct sinfo Student;
    struct SStudentNode *PNextStudent; // special pointer to struct of same type of the node !
}SStudentNode;

// Queue structure
typedef struct St_Queue {
    SStudentNode* front;
    SStudentNode* rear;
} St_Queue;

extern St_Queue* student_queue ;
extern int count;
extern FILE* filePointer;
```

```

// Functions / APIs:

// Function to create a new node / Student
SStudentNode* createNode_student(struct sinfo student_data);

// Function to initialize the queue
St_Queue* createQueue_student();

// Function to add an element to the queue
void enqueue_student(struct sinfo student_data);

// Function to add the student manually to the queue
void add_student_manually();

// Function to add the student from the file to the queue
void add_student_file();

// Function to check the roll number of the student in the queue
int check_roll(int num);

// Function to update the student data in the queue
void up_s();

// Function to find the student with his roll number in the queue
void find_rl();

// Function to find the student with his first name in the queue
void find_fn();

// Function to find the student with his course in the queue
void find_c();

// Function to calculate the total number of students in the queue
void tot_s();

// Function to remove the student manually from the queue
void del_s();

// Function to display single student
void display_student(int rl);

// Function to display the queue (all students)
void show_s();

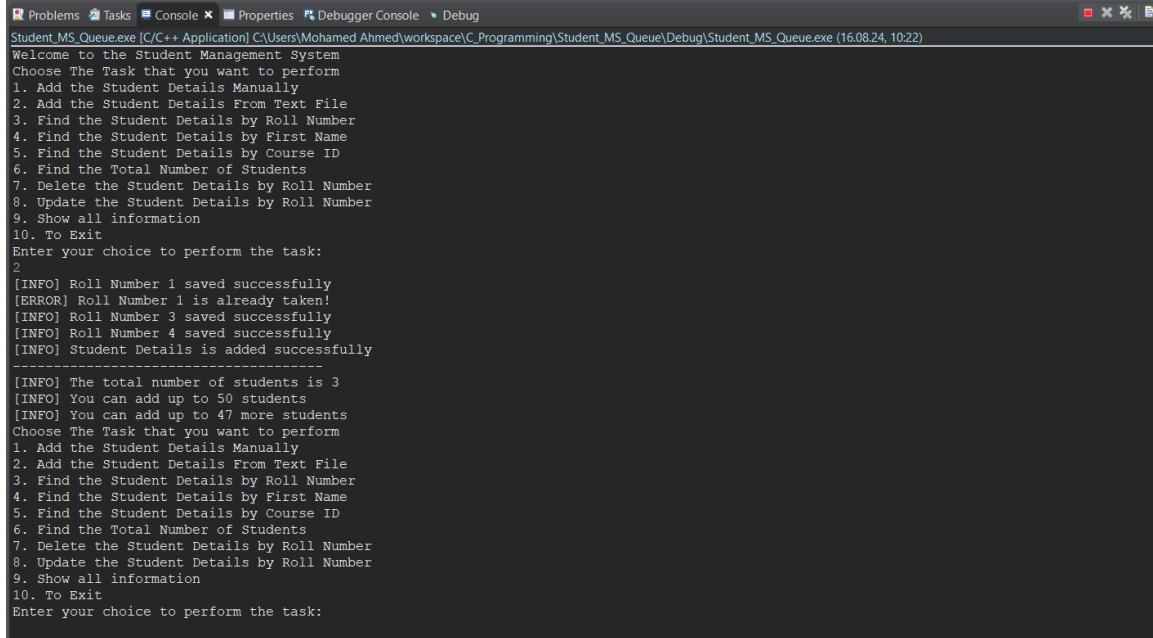
#endif /* QUEUE_LINKEDLIST_H_ */

```

The implementation file “**Queue_LinkedList.c**” is huge and would take a lot of screenshots to cover. Therefore, I would only refer to it here with the following github.com project under this link:
https://github.com/MohamedFathalla91/Master_Embedded_Systems/blob/development/Unit_5_First_Term_Projects/Project2_Student_Management_System/Queue_LinkedList.c

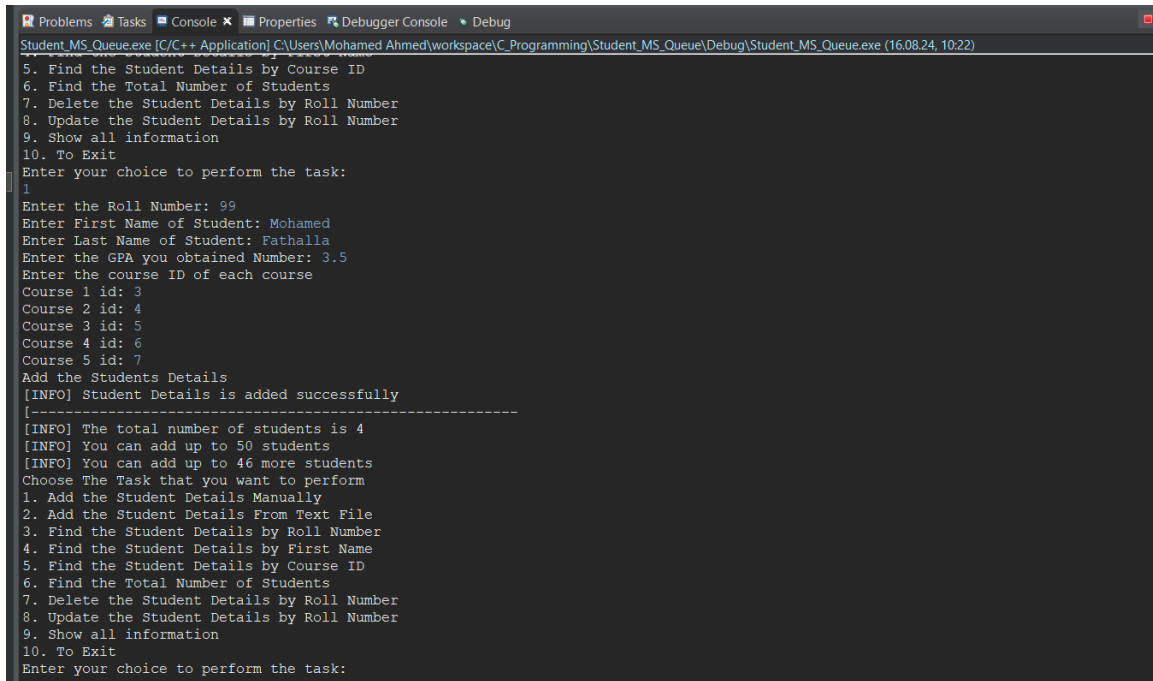
Software Testing Results

Add Students from File:



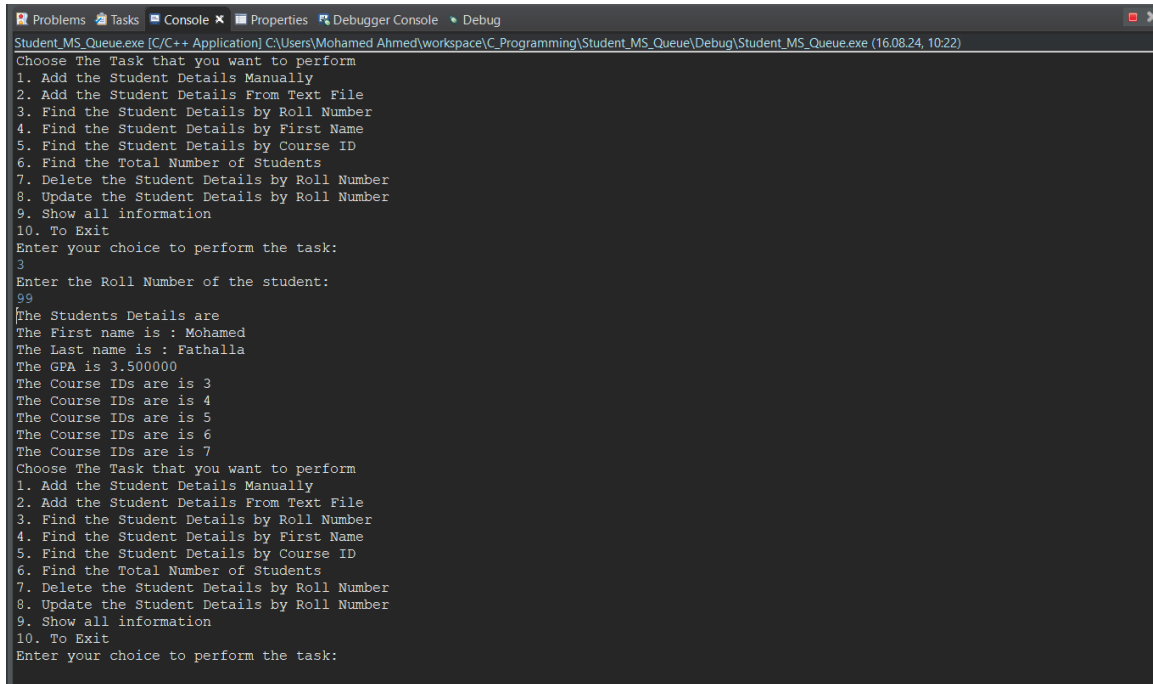
```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
Welcome to the Student Management System
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
2
[INFO] Roll Number 1 saved successfully
[ERROR] Roll Number 1 is already taken!
[INFO] Roll Number 3 saved successfully
[INFO] Roll Number 4 saved successfully
[INFO] Student Details is added successfully
-----
[INFO] The total number of students is 3
[INFO] You can add up to 50 students
[INFO] You can add up to 47 more students
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Add Student Manually:



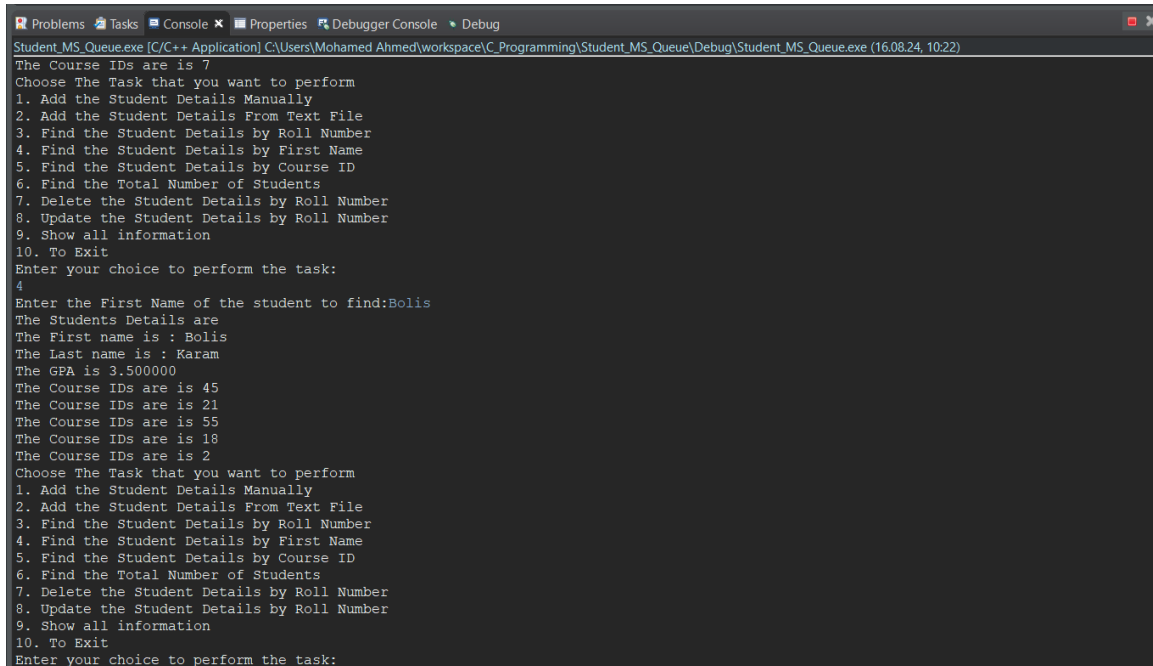
```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
1
Enter the Roll Number: 99
Enter First Name of Student: Mohamed
Enter Last Name of Student: Fathalla
Enter the GPA you obtained Number: 3.5
Enter the course ID of each course
Course 1 id: 3
Course 2 id: 4
Course 3 id: 5
Course 4 id: 6
Course 5 id: 7
Add the Students Details
[INFO] Student Details is added successfully
-----
[INFO] The total number of students is 4
[INFO] You can add up to 50 students
[INFO] You can add up to 46 more students
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Find the student by roll number:



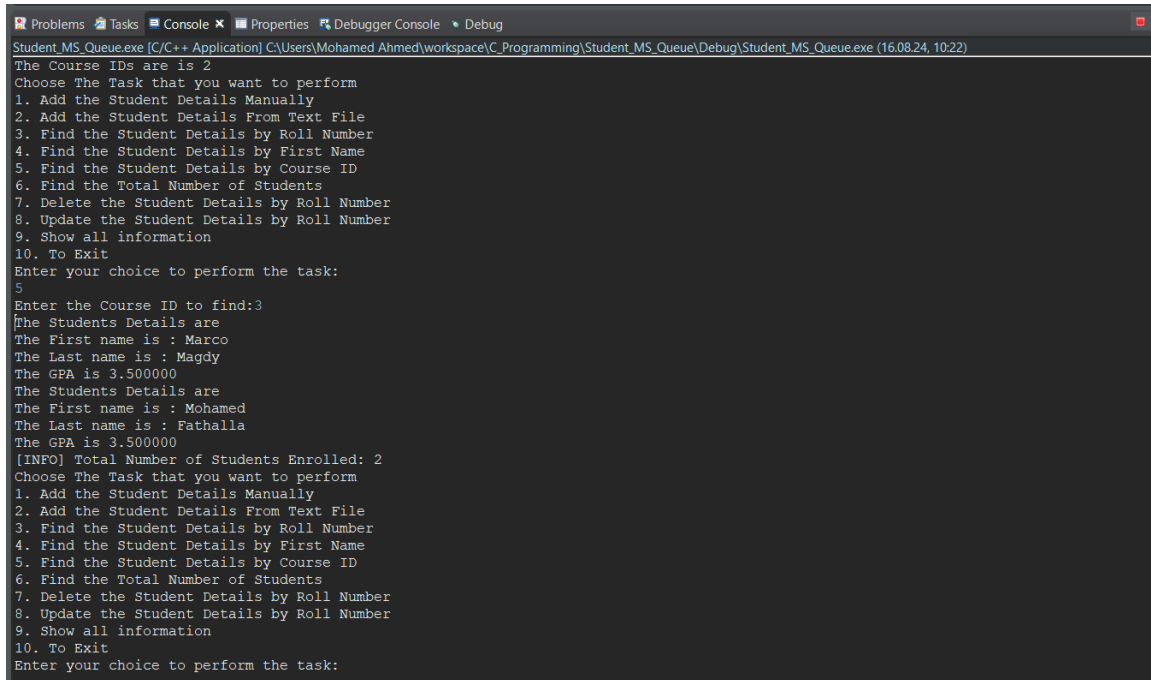
```
Student_MS.Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS.Queue\Debug\Student_MS.Queue.exe (16.08.24, 10:22)
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
3
Enter the Roll Number of the student:
99
The Students Details are
The First name is : Mohamed
The Last name is : Pathalla
The GPA is 3.500000
The Course IDs are is 3
The Course IDs are is 4
The Course IDs are is 5
The Course IDs are is 6
The Course IDs are is 7
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Find the student by the first name:



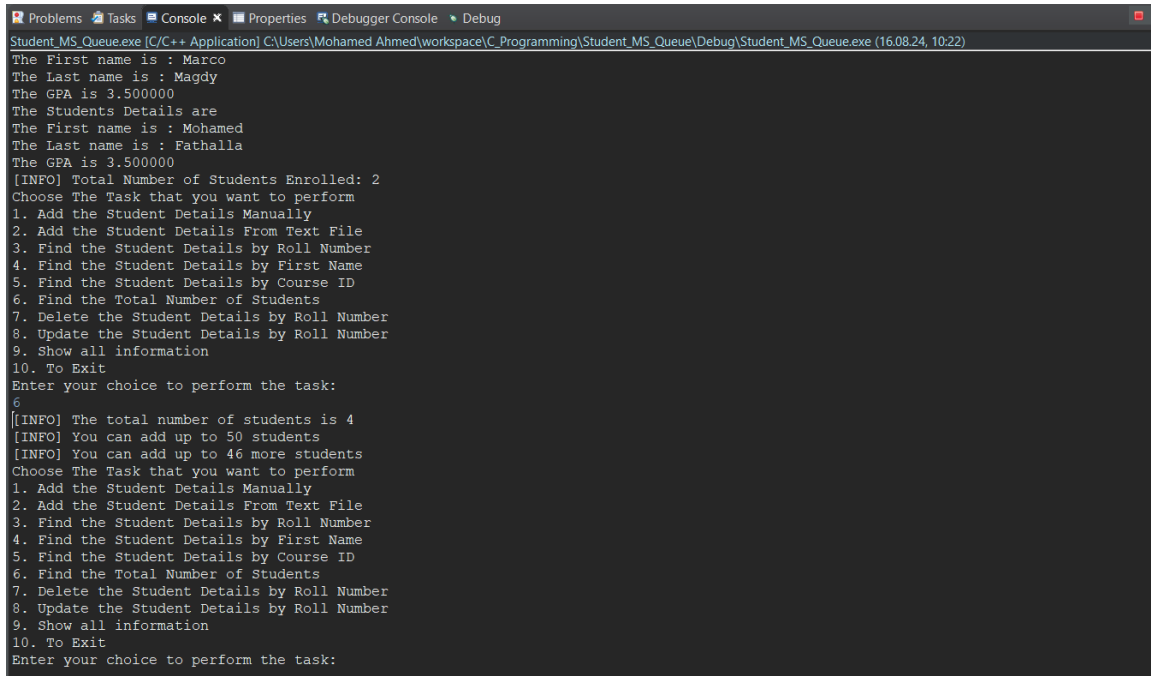
```
Student_MS.Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS.Queue\Debug\Student_MS.Queue.exe (16.08.24, 10:22)
The Course IDs are is 7
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
4
Enter the First Name of the student to find:Bolis
The Students Details are
The First name is : Bolis
The Last name is : Karam
The GPA is 3.500000
The Course IDs are is 45
The Course IDs are is 21
The Course IDs are is 55
The Course IDs are is 18
The Course IDs are is 2
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Find all students with the same course ID:



```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
The Course IDs are is 2
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
5
Enter the Course ID to find:3
The Students Details are
The First name is : Marco
The Last name is : Magdy
The GPA is 3.500000
The Students Details are
The First name is : Mohamed
The Last name is : Fathalla
The GPA is 3.500000
[INFO] Total Number of Students Enrolled: 2
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Calculate the total number of students:



```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
The First name is : Marco
The Last name is : Magdy
The GPA is 3.500000
The Students Details are
The First name is : Mohamed
The Last name is : Fathalla
The GPA is 3.500000
[INFO] Total Number of Students Enrolled: 2
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
6
[INFO] The total number of students is 4
[INFO] You can add up to 50 students
[INFO] You can add up to 46 more students
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
```

Remove single student by his roll number & display all students:

```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
7
Enter the Roll Number which you want to delete:
3
[INFO] This Roll Number 3 is removed successfully!
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
9
The First name is : Marco
The Last name is : Magdy
The GPA is 3.50
The Course IDs are 1
The Course IDs are 2
The Course IDs are 3
The Course IDs are 4
The Course IDs are 5
-----
The First name is : Kerolos
The Last name is : Gamal
The GPA is 3.50
```

```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
9. Show all information
10. To Exit
Enter your choice to perform the task:
9
The First name is : Marco
The Last name is : Magdy
The GPA is 3.50
The Course IDs are 1
The Course IDs are 2
The Course IDs are 3
The Course IDs are 4
The Course IDs are 5
-----
The First name is : Kerolos
The Last name is : Gamal
The GPA is 3.50
The Course IDs are 45
The Course IDs are 2
The Course IDs are 55
The Course IDs are 18
The Course IDs are 46
-----
The First name is : Mohamed
The Last name is : Fathalla
The GPA is 3.50
The Course IDs are 3
The Course IDs are 4
The Course IDs are 5
The Course IDs are 6
The Course IDs are 7
-----
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
```

Update single student and display the all students:

```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
ERROR: Roll Number 3 not found.
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
8
Enter the Roll Number to update the entry:
99
1. first name
2. last name
3. roll no.
4. GPA
2
Enter the new last name: Ahmed
[INFO] UPDATED SUCCESSFULLY.
Choose The Task that you want to perform
1. Add the Student Details Manually
2. Add the Student Details From Text File
3. Find the Student Details by Roll Number
4. Find the Student Details by First Name
5. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
9
The First name is : Marco
The Last name is : Magdy
```

```
Problems Tasks Console Properties Debugger Console Debug
Student_MS_Queue.exe [C/C++ Application] C:\Users\Mohamed Ahmed\workspace\C_Programming\Student_MS_Queue\Debug\Student_MS_Queue.exe (16.08.24, 10:22)
6. Find the Student Details by Course ID
6. Find the Total Number of Students
7. Delete the Student Details by Roll Number
8. Update the Student Details by Roll Number
9. Show all information
10. To Exit
Enter your choice to perform the task:
9
The First name is : Marco
The Last name is : Magdy
The GPA is 3.50
The Course IDs are 1
The Course IDs are 2
The Course IDs are 3
The Course IDs are 4
The Course IDs are 5
-----
The First name is : Kerolos
The Last name is : Gamal
The GPA is 3.50
The Course IDs are 45
The Course IDs are 2
The Course IDs are 55
The Course IDs are 18
The Course IDs are 46
-----
The First name is : Mohamed
The Last name is : Ahmed
The GPA is 3.50
The Course IDs are 3
The Course IDs are 4
The Course IDs are 5
The Course IDs are 6
The Course IDs are 7
-----
Choose The Task that you want to perform
```